# A Unified Framework for Nonlinear Local–Global Latent Dynamics, Koopman Modal Signatures, and Two LFP–BOLD Projects

Framework note

2026-06-10

**Core idea**

LFP, local BOLD, and global BOLD should not be treated as a simple feed-forward input-output chain. They are better understood as different observable readouts of a nonlinear local-global latent system. Koopman eigenfunctions transform this complex nonlinear system into interpretable modal variables, making the global gating state, pre-activation modes, global ROI maps, and loop backbones operational phenomenological modal signatures.

## Contents


**1 Overview** 2

**2 Underlying model: a nonlinear local-global latent system** 2

**3 BOLD is a nonlinear readout, not the mechanism itself** 3

**4 Global-to-local path: global state gates local LFP subprocesses** 3

**5 Why local BOLD–LFP can produce an apparent non-causal IR** 3

**6 From nonlinear system to augmented autonomous system** 4

**7 Koopman modal representation** 4

**8 Definition of pre-activation modes** 5

**9 Koopman phase-lead explanation** 5

**10 Meaning of the global ROI map** 5

**11 Meaning of the loop backbone** 6

**12 Relation to Haken's slaving principle** 7

**13 Relationship to the two projects** 7

13.1 Project 1: Event-related LFP subprocess → global BOLD pattern / BOLD eigenfunction residual . . . 7

13.2 Project 2: Non-causal LFP–BOLD IR / pre-activation . . . 8

**14 Integrated closed-loop interpretation** 8

**15 Interpretation boundaries** 9

# 1 Overview

This note summarizes the current mathematical framework connecting two related projects:

1. **Event-related LFP subprocess → global BOLD pattern / BOLD eigenfunction residual**: how local LFP subprocesses can act as intrinsic perturbations that trigger or predict global BOLD modal innovations.

2. **Non-causal LFP–BOLD impulse responses / pre-activation**: why local BOLD can show apparent pre-activation relative to local LFP, and how this can be interpreted using closed-loop local-global latent dynamics and a Koopman modal representation.

The two projects correspond to two halves of the same closed loop:

$$\underbrace{u_{\text{LFP}} \to x^{(g)}}_{\text{local-to-global: BOLD modal innovation}} \qquad \text{and} \qquad \underbrace{x^{(g)} \to u_{\text{LFP}}}_{\text{global-to-local: pre-activation / gating}}.$$

Together, they describe

$$x^{(g)} \to u_{\text{LFP}} \to x^{(g)},$$

meaning that a global order-parameter-like state gates local hippocampal subprocesses, and local subprocesses in turn perturb or update the global BOLD network state.

# 2 Underlying model: a nonlinear local-global latent system

Let the latent state be decomposed into local and global components:

$$x_t = \begin{bmatrix} x_t^{(l)} \\ x_t^{(g)} \end{bmatrix}. \tag{1}$$

Here:

- $x_t^{(l)}$ is a **local latent state**, such as hippocampal local neural state, local vascular state, or local circuit excitability.

- $x_t^{(g)}$ is a **global latent state / internal environment / global order-parameter-like state**, such as global brain state, neuromodulatory state, brainstem-thalamic-cortical state, global vascular state, or arousal state.

- $u_t$ is a **local LFP subprocess / local neural action**.

A generic nonlinear model can be written as

$$x_{t+1}^{(l)} = F_l\left(x_t^{(l)}, x_t^{(g)}, u_t\right), \tag{2}$$

$$x_{t+1}^{(g)} = F_g\left(x_t^{(g)}, x_t^{(l)}, u_t\right), \tag{3}$$

or more compactly:

$$x_{t+1} = F(x_t, u_t). \tag{4}$$

The key point is not that LFP is an external stimulus. Rather, $u_t$ is treated as an action-like subprocess of the local subsystem. It can be written as an input-like variable for modeling convenience, but in the closed-loop view it is generated internally by the system.

## 3 BOLD is a nonlinear readout, not the mechanism itself

Local BOLD and global BOLD are observable readouts of the latent state:

$$y_t^{(lB)} = h_l\left(x_t^{(l)}, x_t^{(g)}\right), \tag{5}$$

$$y_t^{(gB)} = h_g\left(x_t^{(l)}, x_t^{(g)}\right). \tag{6}$$

Therefore, local BOLD is not simply the output of a local HRF driven by local LFP. It may contain both a local and a global component:

$$y_t^{(lB)} = \text{local component} + \text{global component}. \tag{7}$$

This is crucial: if local BOLD reads out a global latent state before that state gates a future local LFP subprocess, local BOLD can show apparent pre-activation relative to LFP.

## 4 Global-to-local path: global state gates local LFP subprocesses

The central explanation for pre-activation is:

$$x_t^{(g)} \to u_{\mathrm{LFP},t+\Delta}. \tag{8}$$

More explicitly:

$$x_t^{(g)} \to x_{t+\Delta}^{(l)} \to u_{\mathrm{LFP},t+\Delta}. \tag{9}$$

This means that a global latent state / order-parameter-like variable appears first, then modulates the local hippocampal condition, making a local LFP subprocess more likely to occur.

To form a complete closed loop, there may also be a local-to-global update:

$$u_{\mathrm{LFP},t} \to x_{t+\Delta}^{(g)}. \tag{10}$$

The overall loop can therefore be summarized as

$$x^{(g)} \to u_{\mathrm{LFP}} \to x^{(g)}. \tag{11}$$

However, for explaining **pre-activation**, the most important part is the first half:

$$x^{(g)} \to u_{\mathrm{LFP}}. \tag{12}$$

## 5 Why local BOLD–LFP can produce an apparent non-causal IR

The standard feed-forward HRF model assumes:

$$y_{\mathrm{localBOLD}}(t) = \sum_{\tau \geq 0} h(\tau) u_{\mathrm{LFP}}(t-\tau) + \epsilon(t). \tag{13}$$

This treats LFP as an exogenous input and BOLD as a passive output. Under this assumption, if the estimated impulse response has negative-lag pre-activation, it appears non-causal.

In the local-global latent model, a global state $x_t^{(g)}$ can simultaneously:

1. be read out early by local BOLD:
$$x_t^{(g)} \to y_t^{(lB)}, \tag{14}$$

2. gate a future local LFP subprocess:
$$x_t^{(g)} \to u_{\mathrm{LFP},t+\Delta}. \tag{15}$$

Thus the observations may show

$$y_{\text{localBOLD},t} \quad \text{preceding} \quad u_{\text{LFP},t+\Delta}. \tag{16}$$

If one still applies a feed-forward LFP-to-BOLD deconvolution model, the resulting cross-kernel can show negative-lag pre-activation. This does **not** imply

$$y_{\text{localBOLD}} \to u_{\text{LFP}}. \tag{17}$$

Instead, it reflects the shared influence of a latent global state:

$$x^{(g)} \to \begin{cases} y_{\text{localBOLD}}, \\ u_{\text{LFP}}. \end{cases} \tag{18}$$

The apparent non-causality therefore arises when a closed-loop endogenous system is interpreted using an open-loop feed-forward model.

# 6 From nonlinear system to augmented autonomous system

If $u_t$ is generated internally, it can be written as

$$u_t = \pi\left(x_t^{(g)}, x_t^{(l)}, u_{t-1}, u_{t-2}, \dots\right) + \epsilon_t. \tag{19}$$

By augmenting the state with $u_t$ and its history,

$$s_t = \begin{bmatrix} x_t \\ u_t \\ u_{t-1} \\ \vdots \end{bmatrix}, \tag{20}$$

the closed-loop system can be written in autonomous form:

$$s_{t+1} = F_{\text{cl}}(s_t). \tag{21}$$

The readouts can be written as

$$u_t = C_u s_t, \tag{22}$$

$$y_t^{(lB)} = C_l s_t, \tag{23}$$

$$y_t^{(gB)} = C_g s_t. \tag{24}$$

This step is important because it shows that the local LFP subprocess, local BOLD, and global BOLD can all be treated as different readouts of the same autonomous closed-loop system.

# 7 Koopman modal representation

For the augmented autonomous system

$$s_{t+1} = F_{\text{cl}}(s_t), \tag{25}$$

let $a_t$ denote Koopman eigenfunction coordinates. In modal coordinates:

$$a_{t+1} = \Lambda a_t. \tag{26}$$

The observables can be approximated as linear readouts:

$$u_{\text{LFP},t} = C_u a_t, \tag{27}$$

$$y_{\text{localBOLD},t} = C_l a_t, \tag{28}$$

$$y_{\text{globalBOLD},t} = C_g a_t. \tag{29}$$

The value of the Koopman representation is that the original latent dynamics and observation functions may be nonlinear, but in lifted/modal space the dominant modes, phase relationships, and spatial readouts can be analyzed linearly or approximately linearly.

## 8 Definition of pre-activation modes

For an LFP–BOLD anchor pair $(p, q)$:

- $p$ denotes a local LFP band/subprocess, such as PGO, theta, or ripple.
- $q$ denotes a local BOLD ROI or local BOLD readout.

If the empirical cross-kernel $h_{q,p}(\tau)$ shows pre-activation at $\tau < 0$, we can identify Koopman modes that contribute to this negative-lag component.

Define:

$$\mathcal{I}_{\mathrm{pre}}(q, p) = \{i : \text{mode } i \text{ contributes to the negative-lag component of pair } (q, p)\}. \tag{30}$$

These are the **pre-activation modes** for the anchor pair.

Biological interpretation:

> These modes most naturally correspond to a global latent gating state / order-parameter-like variable. This global state is read out by BOLD before it gates future local LFP subprocesses.

Important boundary:

> Pre-activation modes are not causal modes. They are pair-conditioned modal signatures.

## 9 Koopman phase-lead explanation

For mode $i$, let the continuous-time eigenvalue be

$$\nu_i = \alpha_i + \mathrm{i}\omega_i. \tag{31}$$

The complex readout weights for LFP and local BOLD can be written as

$$C_u(i) = |C_u(i)|e^{\mathrm{i}\phi_{u,i}}, \tag{32}$$
$$C_l(i) = |C_l(i)|e^{\mathrm{i}\phi_{l,i}}. \tag{33}$$

If the BOLD readout phase leads the LFP readout phase, this mode contributes to apparent pre-activation in the LFP–BOLD cross-kernel. A phase-derived time difference can be written as

$$\Delta t_{lB-u,i} = \frac{\mathrm{wrap}(\phi_{l,i} - \phi_{u,i})}{\omega_i}, \tag{34}$$

with the exact sign depending on the phase convention. Conceptually:

$$\text{BOLD phase leads LFP phase} \quad \Rightarrow \quad \text{negative-lag contribution in empirical IR.} \tag{35}$$

## 10 Meaning of the global ROI map

For the pre-activation modes of an anchor pair $(q, p)$, define the global ROI map as

$$M_{q,p}(r) = \sum_{i \in \mathcal{I}_{\mathrm{pre}}(q,p)} s_i |C_g(r, i)|, \tag{36}$$

where:

- $r$ is a global BOLD ROI.
- $C_g(r, i)$ is the global BOLD readout weight of mode $i$ at ROI $r$.

- $s_i$ is the strength with which mode $i$ contributes to the pair-specific pre-activation, for example negative-lag area, mode amplitude, or pairwise mode weight.

Interpretation:

> The global ROI map is the spatial footprint of the pre-activation modes in whole-brain BOLD ROI space.

In the closed-loop / slaving-principle interpretation, it further corresponds to:

> the BOLD-observable footprint of the global latent state / order-parameter-like variable that gates the local LFP subprocess.

It is **not** a causal map of BOLD ROIs driving LFP. It is a map of which ROIs jointly express the global gating state.

## 11 Meaning of the loop backbone

For the same pre-activation modes, each ROI readout has both amplitude and phase:

$$C_g(r,i) = |C_g(r,i)|e^{\mathrm{i}\phi_{r,i}}. \tag{37}$$

The global ROI map uses the amplitude:

$$|C_g(r,i)|, \tag{38}$$

whereas the loop backbone uses the phase:

$$\phi_{r,i}. \tag{39}$$

Thus:

> The global ROI map describes where the global state is expressed. The loop backbone describes how that global state is phase-expressed across ROIs.

For each mode, one may define a phase-based adjacency:

$$A_i(r_1,r_2) = 1 \quad \text{if ROI } r_1 \text{ phase-leads ROI } r_2 \quad \text{within mode } i. \tag{40}$$

Aggregating over pre-activation modes gives

$$A_{q,p}^{\text{backbone}}(r_1,r_2) = \sum_{i\in\mathcal{I}_{\text{pre}}(q,p)} s_i A_i(r_1,r_2). \tag{41}$$

The loop backbone is interpreted as:

> the phase-ordered expression, or phase portrait, of the global gating state in ROI space.

It is a candidate biological representation / phenomenological modal signature. It is not the anatomical feedback loop itself, and it does not by itself establish interventional causality.

## 12 Relation to Haken's slaving principle

This framework can be related to Haken's slaving principle:

$$\text{distributed brain regions} \Rightarrow \text{global order parameter} \Rightarrow \text{local fast subprocess.} \tag{42}$$

In the current setting:

$$\text{distributed ROI-level activity} \Rightarrow x^{(g)} \Rightarrow u_{\text{LFP}}. \tag{43}$$

Here:

- $x^{(g)}$ is a global order-parameter-like state / slaving variable.
- The global ROI map is the spatial BOLD footprint of $x^{(g)}$.
- The loop backbone is the phase portrait of $x^{(g)}$ across ROIs.
- The local LFP subprocess is the fast local process gated or enslaved by $x^{(g)}$.

This also forms a kind of circular causality:

$$\text{distributed global state} \to \text{local LFP subprocess} \to \text{global state update.} \tag{44}$$

## 13 Relationship to the two projects

### 13.1 Project 1: Event-related LFP subprocess → global BOLD pattern / BOLD eigenfunction residual

This project corresponds to the local-to-global path:

$$u_{\text{LFP}} \to x^{(g)} \to y_{\text{globalBOLD}}. \tag{45}$$

Operationally, one first builds a BOLD-only Koopman model:

$$b_{k+1} = \Lambda_B b_k + \eta_k^B, \tag{46}$$

where the BOLD modal innovation / BOLD eigenfunction residual is

$$\eta_k^B = b_{k+1} - \Lambda_B b_k. \tag{47}$$

The high-rate LFP subprocesses are then converted to the BOLD time scale as density variables:

$$D_r^{\text{LFP}}[k] = \text{density / occupancy of LFP subprocess } r \text{ in BOLD window } k. \tag{48}$$

The core predictive model is

$$\eta_{m,k}^B = \sum_r \sum_\ell \beta_{m,r,\ell} D_r^{\text{LFP}}[k-\ell] + \epsilon_{m,k}. \tag{49}$$

If $D_r^{\text{LFP}}$ predicts future $\eta^B$, then:

> the local LFP subprocess acts as an intrinsic perturbation that triggers or predicts global BOLD modal innovations.

Important boundary:

> The BOLD residual itself is not the LFP subprocess activation. It is an input-like deviation of BOLD modal dynamics. It supports the LFP-subprocess-as-intrinsic-perturbation interpretation only if it is predicted by LFP subprocess density.

### 13.2 Project 2: Non-causal LFP–BOLD IR / pre-activation

This project corresponds to the global-to-local path:

$$x^{(g)} \to u_{\mathrm{LFP}}. \tag{50}$$

The proposed sequence is:

1. A global latent state $x^{(g)}$ appears first.
2. Local BOLD reads out a local/global mixture that includes $x^{(g)}$.
3. The same $x^{(g)}$ gates a future local LFP subprocess.
4. Therefore a feed-forward LFP-to-local-BOLD IR can show apparent pre-activation.

In the Koopman analysis:

- The anchor pair $(q, p)$ selects the pre-activation modes $\mathcal{I}_{\mathrm{pre}}(q, p)$.
- The global ROI map $M_{q,p}(r)$ is the spatial footprint of these modes.
- The loop backbone $A_{q,p}^{\mathrm{backbone}}$ is their phase-ordered expression across ROIs.

This project argues:

> BOLD pre-activation is not reverse neurovascular causality. It is a phenomenological modal signature of a global gating state that precedes and organizes local LFP subprocesses.

## 14 Integrated closed-loop interpretation

The two projects correspond to two halves of the same closed loop:

| **Path** | **Project** | **Interpretation** |
|---|---|---|
| $u_{\mathrm{LFP}} \to x^{(g)}$ | Event-related LFP subprocess $\to$ BOLD innovation | Local subprocess perturbs / updates global BOLD modal state |
| $x^{(g)} \to u_{\mathrm{LFP}}$ | Non-causal IR / pre-activation | Global gating state precedes and gates local LFP subprocess |

Together:

$$x^{(g)} \to u_{\mathrm{LFP}} \to x^{(g)}. \tag{51}$$

In words:

> A global order-parameter-like state gates local hippocampal subprocesses, and local subprocesses in turn perturb or update the global BOLD network state.

This closed-loop local-global self-organization is the shared mechanistic interpretation of the two projects.

## 15 Interpretation boundaries

The framework should be stated carefully:

- A **pre-activation mode** is not a causal mode; it is a pair-conditioned modal signature.
- A **global ROI map** is not a causal feedback map; it is the spatial footprint of pre-activation modes.
- A **loop backbone** is not an anatomical feedback loop; it is the phase-ordered expression of a global gating state in ROI space.
- The whole interpretation is phenomenological but mechanistically informative.

A concise manuscript-friendly formulation is:

> Pre-activation modes identify a global order-parameter-like state associated with an LFP–BOLD anchor pair. The global ROI map describes where this state is expressed in BOLD space, while the loop backbone describes how that state is phase-expressed across ROIs. These objects are phenomenological modal signatures of a latent feedback/slaving process, rather than direct anatomical or interventional feedback pathways.